Project Proposal: Home Energy Coach

A Personalized Green Habit Recommender for Students and Renters in Shared Flats

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High Integrity Systems – Winter Semester 2026/27
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***Abstract*—This proposal outlines the design and development plan for *Home Energy Coach*, a Personalized Green Habit Recommender developed as part of the High Integrity Systems Project at Frankfurt University of Applied Sciences (WS 2026/27). The application targets students and renters in shared flats (WGs), who cannot change their building's insulation or heating system but can change their daily habits. Users complete a short energy profile, receive a ranked feed of habits that fit their flat and schedule, commit to one small goal at a time and are prompted at the right moment. The signature feature is the *Green Hours planner*: using live generation data of the German electricity market from the Bundesnetzagentur's SMARD platform, it tells each user the lowest-CO_2 hours for laundry, dishwasher and charging, fitted to the hours they are actually at home. The native iOS app (SwiftUI) talks to a FastAPI back-end, which coordinates the application logic and delegates all data analysis – the hourly CO_2 forecast, recommendation ranking, association-rule mining, user clustering and the evaluation of nudges – to an R analytics service exposed through the `plumber` package. User data is persisted in PostgreSQL and the server side is containerised with Docker. In line with the course's green software requirement, the energy and resource consumption of the application itself is measured from the first prototype on, reduced in every sprint and shown to users in the app. The project follows Scrum over five sprints with rotating Scrum Master and Product Owner roles, and all progress is tracked in Jira.**

I. Introduction

Private households account for a substantial share of Germany's final energy consumption, and within a household more than two thirds of that energy goes into space heating, followed at a distance by hot water, cooking and electrical appliances [6]. Most of the advice households receive therefore targets the building: insulate, replace the boiler, install a heat pump. Students and other renters – and in particular the residents of shared flats (Wohngemeinschaften, WGs) – cannot act on any of it. They do not own the building, rarely choose the heating system or the electricity contract, and often never see their own consumption because the flat receives a single bill. What they *can* change is their daily behaviour: how rooms are heated and ventilated, how long showers last, which devices stay on standby and, increasingly important, *when* appliances run.

The timing aspect is what makes this project more than a list of tips. The CO_2 intensity of German electricity varies strongly within a day because the share of wind and solar generation changes from hour to hour. The Bundesnetzagentur publishes actual and forecast generation per energy source and the actual consumption for Germany on its SMARD platform, free of charge and in machine-readable form [3]. From this data an hourly CO_2 intensity can be computed, and a washing-machine load, a dishwasher run or the charging of a laptop can be shifted into the cleanest hours – provided the user is at home then. This is the idea behind the *Green Hours planner*, the signature feature of Home Energy Coach.

Information alone, however, rarely changes behaviour; the gap between good intentions and action is well documented. Home Energy Coach is therefore designed as a habit-building system rather than an information app. It follows the Fogg Behavior Model, in which a behaviour occurs only when motivation, ability and a prompt come together at the same moment [9], and the habit loop of cue, routine and reward. Users commit to one small, concrete goal at a time – an implementation intention of the form "when X, I do Y" [10] – are prompted exactly when the green window opens, and see the saved CO_2 and their streak immediately afterwards. Behaviour that is repeated in the same context for several weeks becomes automatic [11]; the app then fades its prompts and proposes the next habit.

This proposal describes the planned development of Home Energy Coach as the team's realisation of the course topic *Personalized Green Habit Recommender*. The system consists of a native iOS app built with SwiftUI (Section III-A), a FastAPI back-end and an R analytics service that performs all data analysis and data mining tasks and is accessed through the `plumber` package. Development follows Scrum over five sprints, with all work tracked in Jira and source code managed collaboratively on GitHub. A particular advantage of the chosen domain is that the team itself belongs to the target group, so the app can be piloted on its own developers in the final sprint.

A. Objectives

1) Build a platform that recommends personalised home-energy habits to students and renters and prompts them at the right moment, including the greenest hours of the day derived from live German grid data.
2) Use R [1] for all data analysis and data mining tasks: the hourly CO_2 forecast, the recommendation ranking, association rules, clustering and the evaluation of nudges.

TABLE I
SCORING OF THE SHORTLISTED DOMAINS (1–5 PER CRITERION)

Candidate *weight*	Real data 30 %	Habit loop 25 %	Screens 15 %	Doable 15 %	Self-test 15 %	Total
Home Energy Coach	5	5	4	4	5	**4.7**
Food Waste Coach	3	5	4	4	5	4.1
Campus Green Challenge	3	4	4	4	5	3.9
Green Grocery Coach	4	4	4	2	4	3.7
Green Commute Coach	4	4	3	4	3	3.7
Digital Footprint Coach	3	3	4	4	4	3.5

3) Deliver at least eight distinct user interfaces beyond standard CRUD and authentication screens; eleven are planned (Section V).
4) Measure the energy and resource consumption of the application itself from the first prototype on, and reduce it in every subsequent sprint.
5) Follow Scrum [2] with rotating Scrum Master and Product Owner roles, with all progress tracked in Jira.

B. Topic Selection

The course fixes the topic but leaves the application domain open. The team shortlisted six candidate domains and scored them from 1 to 5 against five criteria derived from the course brief: availability of real data (weight 30 %), fit with a cue–routine–reward habit loop (25 %), number of meaningful screens (15 %), feasibility within one semester (15 %) and whether the team can test the app on itself (15 %). Table I shows the result. Home Energy Coach scored highest, mainly because of the free, high-resolution SMARD data and the natural habit loop around appliance timing. The campus-challenge idea survives as the WG and campus challenge feature inside the winner.

C. Target Users

The primary persona is a student in a shared flat: for example *Lena*, 24, a master's student in a three-person WG in Frankfurt. The flat has gas central heating with thermostatic valves in every room, a shared washing machine and one electricity bill for all three flatmates, so Lena has never seen her own kilowatt-hours. She is motivated but busy and will only follow advice that fits her timetable. The secondary persona is a renter outside a shared flat, for example a young professional in a one-room flat who pays their own bills and is interested in the Green Hours planner and the heating coaching, but not in the group features. Both personas, together with scenarios and user stories, are refined in Sprint 1.

D. Problem Statement

The aim of the project is to turn public energy data and a short user profile into habits that people actually keep. The following challenges have to be addressed explicitly:

- **No consumption measurements.** Most residents of shared flats do not know their own kWh: there is one bill per flat, usually no smart meter and no submetering. Savings therefore have to be estimated from the profile and from public statistics, with optional meter readings as a refinement.
- **Generic advice fails.** Lowering the thermostat is useless advice for a flat with district heating and no valves; a laundry tip is pointless without a washing machine. Recommendations must respect flat type, heating type, available appliances and the user's weekly schedule.
- **Awareness is not behaviour.** Dashboards alone do not change routines. The app has to provide prompts at the right moment, small commitments, immediate feedback and a visible reward, and it has to recognise when a habit has formed so that prompts can fade.
- **The same kWh is not always equally green.** Whether a load runs at noon or at 19:00 can change its CO_2 footprint considerably. The app needs current and forecast grid data and must match green windows to the hours the user is at home.
- **Savings in a shared flat are social.** Heating and hot water are shared; one flatmate alone cannot save much. The app needs group goals, a WG agreement and challenges – without turning into surveillance of flatmates.
- **Cold start.** There are no real users at the start of the project. The recommender has to be developed and evaluated on synthetic users derived from public statistics, a survey and a pilot with the team itself.
- **The app's own footprint.** An app that nudges people to save energy while wasting energy on its own servers would contradict its purpose. Its consumption has to be measured and reduced as part of the project.

E. Scope

In scope. A native iOS app with the eleven screens of Section V; a FastAPI back-end with PostgreSQL; an R analytics service with the pipelines of Section II-E; the Green Hours planner for the German electricity grid, based on SMARD data; a catalogue of 22 home-energy habits covering heating, ventilation, hot water, appliances and standby; WG groups, agreements and challenges; the measurement and reduction of the app's own footprint; and a two-week pilot with the team in Sprint 5.

Out of scope. Android and web clients; automatic control of appliances, smart plugs or thermostats; automatic integration of smart meters or utility accounts (meter readings are entered manually); electricity grids outside Germany; other areas of the course topic such as transport, food and shopping habits; tariff switching and payments; and a public App Store release – the app reaches pilot users through TestFlight only. These limits keep the project within the available 1 800 team hours; an Android client and smart-plug integration are noted as future work.

F. Example Scenario

Lena's laundry illustrates the intended five-week path from awareness to habit. In week 0 the onboarding quiz tells her that her WG's laundry causes roughly 4 kg CO_2 per month. In week 1 she commits to a single small goal: one load in

TABLE II
SPRINT TIMELINE

#	Dates	Focus
1	26 Oct – 6 Nov	Personas, scenarios and user stories; Jira board and team contract; synthetic-data plan; first SMARD and UBA downloads
2	9 – 20 Nov	Wireframes (Figma), system architecture, UML models, user survey
3	23 Nov – 4 Dec	Back-end and database, R service, SMARD pipeline, first screens
4	7 – 18 Dec	Recommender, Green Hours planner, mockups, green baseline measurement
5	4 – 25 Jan	Pilot with the team, optimisation and re-measurement, final report and delivery

green hours on Saturday. On Saturday at 11:30 the app alerts her that a green window opens from 12:00 to 15:00; she starts the machine at noon and sees $-0.4\,\mathrm{kg}$ CO_2, a streak of one and her WG at rank 2 in the challenge. In weeks 2–4 the goal grows and her flatmates join. From week 5 she runs the machine in green hours without being asked; the prompts fade and the app proposes the next habit, lowering the thermostat by 1 °C. All figures are illustrative; the real ones come from SMARD and the user's profile.

II. METHODOLOGY

A. Development Process

The project follows the Scrum framework and is divided into the five sprints prescribed by the course: four two-week sprints and a final three-week sprint, starting on 26 October 2026 and ending with the final delivery on 25 January 2027 (Table II). The practical classes in the first week of each sprint are used for cumulative project work; in the second week the team presents a summary of the completed work. In addition, the team meets the supervisor in the weekly online project meeting. Every sprint contains Sprint Planning, daily stand-ups, a Sprint Review and a Sprint Retrospective; evidence of all four is documented on the Jira board. The course corresponds to 10 ECTS, i.e. about 300 working hours per member or roughly 1 800 team hours for the six members over the 13 project weeks, which the sprint backlogs are sized against.

All tasks are tracked in Jira with the boards Product Backlog, To Do / In Progress / Done, Artifacts, Daily SCRUM and Team Contract; the instructor has access to the project. The team contract defines the expectations for team meetings, team and individual responsibilities, and the strategy for handling impediments and conflicts.

- **Jira project:** mazba09.atlassian.net/jira/software/projects/HEC
- **GitHub repository:** github.com/Asifnewaz/Personalized_Green_habit_Recommender

The Scrum Master (Team Captain) and the Product Owner rotate as shown in Table III. Because the team has six members but the course has only five sprints, the three-week Sprint 5 is split into two role periods, 5a and 5b, of about ten days each. In this way every member holds each role exactly once, and nobody holds both roles in the same period. The split of Sprint 5 is subject to confirmation by the supervisor. The Product Owner maintains and prioritises the product backlog and decides what the sprint delivers; the Scrum Master facilitates the Scrum events, keeps the board up to date and removes impediments.

TABLE III
ROTATION OF THE SCRUM ROLES

Sprint	Dates	Scrum Master	Product Owner
1	26 Oct – 6 Nov	Mazba Uddin	Asif Newaz
2	9 – 20 Nov	Madhusudan Ambulkar	Muhammad Saad Zafar
3	23 Nov – 4 Dec	Md. Rabiul Hasan Tanvir	Nazmul Alam
4	7 – 18 Dec	Nazmul Alam	Madhusudan Ambulkar
5a	4 – 14 Jan	Muhammad Saad Zafar	Mazba Uddin
5b	15 – 25 Jan	Asif Newaz	Md. Rabiul Hasan Tanvir

B. Team Responsibilities

Besides the rotating Scrum roles, the work is divided into technical and organisational areas (Table IV). An area owner is accountable for the quality and timely delivery of the area and is the first point of contact for it, but does not do all of its work alone. A member can own or contribute to several areas, and an area can have several owners; the R analytics, the largest and most specialised area, has at least two, so that the knowledge does not depend on a single person. Every member writes code and reviews other members' pull requests. Members are listed by surname; the assignment is also recorded in the team contract in Jira.

C. Data Collection and Integration

The application combines three kinds of data: open data about the German electricity system, statistics about typical households, and the data generated by its own users. Table V lists the sources and their use.

SMARD provides the actual generation per energy source (lignite, hard coal, gas, wind, solar, hydro, biomass, …), the day-ahead generation forecast and the actual consumption for the German market area in hourly and quarter-hourly resolution. The data goes back to 2015, is free of charge and is available through JSON endpoints, which can be queried from R either directly with `httr2` or through the CRAN package `bunddev`, which wraps the SMARD endpoint with caching and rate limiting [8]. The hourly CO_2 intensity of the grid is computed from the generation mix and the per-source emission factors; the yearly average published by the Umweltbundesamt – 344 g CO_2/kWh for 2025 (estimate), 353 g for 2024 [4] – serves as the default conversion wherever no hourly value is needed, e.g. for yearly savings. The corresponding UBA factors for natural gas and district heating are used for heating habits.

Household statistics give the app a baseline before a user has produced any data. The Stromspiegel 2025 of co2online, based

TABLE IV
AREAS OF RESPONSIBILITY

Area	Main responsibilities	Members
R analytics	SMARD pipeline and Green Hours forecast, recommender, association rules, clustering, evaluation; `testthat` tests	Ambulkar, Tanvir
Back-end and database	FastAPI services, API contract, authentication, PostgreSQL schema, scheduler, R bridge; `pytest` tests	Uddin, Zafar
iOS app	SwiftUI screens, local and push notifications, offline cache, accessibility; XCTest and XCUITest	Newaz
UX and design	Personas, scenarios, survey, Figma wireframes and mockups, usability tests in the pilot	Newaz, Tanvir
Architecture and modelling	Architecture document and its justification; use case, class, activity and sequence diagrams	Zafar, Newaz, Ambulkar
Data and synthetic users	Open-data downloads (UBA, Stromspiegel, REFIT), habit catalogue figures, synthetic users, survey analysis	Ambulkar, Alam
Green software	Baseline and per-sprint measurements, optimisation backlog, data for the "How green is this app?" screen	Uddin, Alam
DevOps and quality	Docker Compose, GitHub Actions, test strategy, code-review rules, Definition of Done	Zafar, Newaz
Documentation	Proposal, user documentation, final report, Artifacts board in Jira	Alam, Tanvir

TABLE V
DATA SOURCES

Need	Source	Use in the app
Grid CO_2 by hour	SMARD (Bundesnetzagentur): generation by source and consumption, hourly, data since 2015, free [3]	Green Hours forecast in R
kWh $\rightarrow$ CO_2	UBA emission factor of the German electricity mix: 344 g CO_2/kWh (2025, estimate) [4]	Default conversion for savings
"Similar flats" baseline	Stromspiegel 2025 (co2online, about 57 000 electricity bills) [5]	Onboarding comparison, savings estimates
Split by end use	UBA household energy data (space heating $>2/3$) [6]	Which habits matter most
When appliances run	REFIT: 20 UK homes, 2 years, appliance-level loads [7]	Realistic synthetic users
Habit adoption	Synthetic users, survey, Sprint 5 pilot	Training and evaluation of the recommender

on about 57 000 electricity bills, provides typical consumption by household size and building type [5]; it powers the "similar flats" comparison during onboarding and the estimate of what a habit saves. The UBA statistics on the energy consumption of private households provide the split by end use – space heating, hot water, cooking, appliances – and thus the weighting of which habits matter most [6]. The REFIT dataset, appliance-level load measurements of 20 UK households over two years [7], provides realistic patterns of when washing machines, dishwashers and other appliances actually run; it is used to generate plausible synthetic users for the cold-start phase.

User-generated data consists of the onboarding quiz, commitments, check-ins, dismissed tips, optional meter readings and notification responses. A short survey among fellow students in Sprint 2 and the pilot with the team in Sprint 5 provide the first real habit-adoption data. An exploratory data analysis in R during Sprints 1 and 2 examines consumption distributions, the daily and seasonal shape of the CO_2 intensity curve and typical appliance run times before any modelling decision is made.

D. Data Preprocessing Pipeline

All preprocessing is implemented in R (`dplyr`, `tidyr`, `lubridate`, `httr2`, `jsonlite`) inside the analytics service, so that the same code serves the exploratory phase and the production pipeline:

- **SMARD ingestion.** A nightly job downloads the actual generation per source for the past day and the generation forecast for the next day, resamples quarter-hourly values to hours, aligns timestamps to Europe/Berlin including daylight-saving changes, fills short gaps (carry-forward up to two hours, longer gaps are flagged) and computes the hourly CO_2 intensity as the emission-factor-weighted mean of the generation mix. The result is stored in PostgreSQL; the last successful download is kept as a fallback snapshot.
- **Profile normalisation.** The quiz answers are converted into a feature vector: flat size, number of flatmates, heating and hot-water type, available appliances, and the weekly home schedule as a 24×7 availability matrix. Consumption of shared flats is expressed per person so that WGs of different sizes are comparable.
- **Habit catalogue.** The 22 habits are stored with their type (timing, reduction, one-time), the estimated saving per execution in kWh, CO_2 and euro, an ease score, prerequisites (e.g. requires a washing machine, requires an own thermostat) and the app tool that supports them (Table VI).
- **Synthetic users.** For the cold start, profiles are drawn from the Stromspiegel consumption bands and household sizes, the end-use split from the UBA data and appliance run times from REFIT; commitment and check-in histories over several weeks are simulated for them and later replaced by real check-ins from the survey and the pilot.
- **Quality control and privacy.** Duplicate check-ins are removed and implausible meter readings are flagged; the analytics service works with pseudonymous user IDs only, and users can export or delete their data from the privacy screen.

E. Data Mining and Recommendation Models (R)

The data processing, modelling pipelines, and execution workflows run in an independent R service connected to the FastAPI back-end through `plumber`. This analytics pipeline

ingests external grid data, generates personalised recommendations, identifies habit connections, segments users, and runs statistical evaluations.

1) **Green Hours Forecast Pipeline**
   - **Model / Algorithm:** Weighted-mean emission factor calculation and time-series forward-fill interpolation.
   - **Packages Used:** `httr2` or `bunddev` (API data fetching), `dplyr` (data manipulation), and `lubridate` (hourly timestamp alignment to Europe/Berlin).
   - **Pipeline:** The service fetches generation mixes and day-ahead forecasts from the SMARD platform. It resamples quarter-hourly entries to hourly intervals, fills gaps up to two hours via carry-forward, and computes the hourly carbon intensity ($g\,CO_2$/kWh) using official UBA emission factors. The forecast is matched against the user's home availability schedule to identify the cleanest hours for flexible appliances like washing machines and dishwashers.
2) **Recommendation Ranking Pipeline**
   - **Model / Algorithm:** Multi-attribute utility heuristic (for cold start) and Item-Based Collaborative Filtering (IBCF).
   - **Packages Used:** `recommenderlab` (collaborative filtering and rating matrices) and `dplyr` (heuristic scoring).
   - **Pipeline:** During onboarding, candidate habits are filtered by user prerequisites (such as heating type and appliance availability) and scored using a utility heuristic based on estimated CO_2 savings, ease, and schedule fit. Once interaction histories and survey records accumulate, an IBCF model calculates item similarities over user completion matrices to recommend habits that succeeded for similar profiles.
3) **Habit Stacking Association Pipeline**
   - **Model / Algorithm:** Apriori association rule mining algorithm.
   - **Packages Used:** `arules` (mining frequent itemsets and association rules).
   - **Pipeline:** Check-in logs are grouped into routine-based transactions. The Apriori algorithm identifies recurring habit combinations using configured support, confidence, and lift thresholds. When a user consistently maintains one habit (e.g., *Stoßlüften*), rules with a lift greater than 1 trigger nudges for a linked habit (e.g., lowering the thermostat).
4) **User Persona Clustering Pipeline**
   - **Model / Algorithm:** k-means clustering (with k-medoids / `pam` as an outlier-resistant alternative).
   - **Packages Used:** `stats` (base `kmeans`), `cluster` (medoid clustering), and `factoextra` (silhouette evaluation).
   - **Pipeline:** Normalised user feature vectors are built from daytime home availability, completion rates across habit types, and notification dismissal rates. Users are partitioned into distinct behavioural clusters (e.g., daytime vs. weekend savers), with the optimal cluster count determined by silhouette analysis. Centroids help tailor default nudges for newly onboarded users.
5) **Evaluation and Nudge Testing Pipeline**
   - **Model / Algorithm:** Cross-validation evaluation metrics (Precision@k, Recall@k, MAE) and paired hypothesis testing (paired t-test / Wilcoxon signed-rank test).
   - **Packages Used:** `recommenderlab` (recommender accuracy benchmarks) and `stats` (hypothesis tests).
   - **Pipeline:** Recommender accuracy is validated on synthetic and survey data splits using 5-fold cross-validation. To evaluate whether nudges produce lasting habits, reminders are faded after an initial consistency period. Paired tests compare check-in adherence before and after fading to verify that behaviour persists without prompts.
6) **API Execution and Green Optimisation Pipeline**
   - **Model / Algorithm:** Scheduled batch processing with algorithmic profiling via `bench` and `profvis`.
   - **Packages Used:** `plumber` (REST API generation), `DBI` and `RPostgres` (database communication), and `bench` / `profvis` (performance and green code profiling).
   - **Pipeline:** Endpoints serve fast lookup queries, while data-heavy jobs (SMARD downloads, clustering, matrix recomputation) execute as scheduled nightly batch runs to keep server CPU usage low. All R functions are profiled across sprints using `bench` and `profvis` to measure and reduce execution runtime and memory overhead in line with green computing goals.

F. Behaviour Change and Personalisation Design

Personalisation happens at three levels. On day one, the *profile* from the onboarding quiz filters the habit catalogue: only habits whose prerequisites the flat fulfils are shown. After one to two weeks, the user's *behaviour* – check-ins, dismissals, response times to prompts – assigns them to a persona cluster, and tips that worked for similar users are preferred (Section II-E). On an ongoing basis, *feedback* adjusts the feed: dismissed tips fade out and mastered habits unlock the next step. Personalisation also applies to time: a user who is at home from 18:00 to 22:00 is told that 19:00 is their greenest slot, not that noon would have been better.

The 22 habits fall into three types that need three different tools (Table VI). Timing habits are driven by the real-time Green Hours prompt; reduction habits run through the habit loop of commitment, check-in and streak; one-time actions are a checklist whose lifetime saving is credited once. Every

TABLE VI
HABIT CATALOGUE BY TYPE

Type	#	Examples	App tool
Timing habits	4	Laundry, dishwasher, charging and batch cooking in green hours	Real-time Green Hours prompt
Reduction habits	12	Heating $-1\,^{\circ}\mathrm{C}$, Stoßlüften, shorter showers, $30\,^{\circ}\mathrm{C}$ wash, standby off	Habit loop: commit, check in, streak
One-time actions	6	LEDs, smart plug, bleeding radiators, green electricity tariff, WG agreement	Checklist with lifetime saving

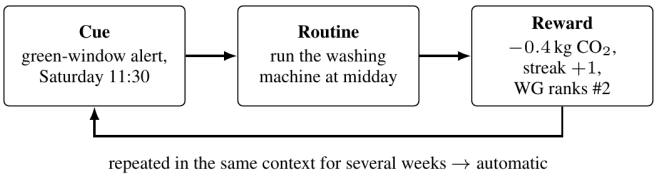


Fig. 1. The habit loop on which every screen of the app is based.

screen maps to one mechanism of the habit loop shown in Fig. 1: the dashboard is feedback, the commitment is an implementation intention, the notification is the prompt, and the streak and the WG ranking are the reward.

Nudging raises ethical questions, and the course brief explicitly asks for ethical AI and transparency. Home Energy Coach therefore uses only opt-in nudges: users choose their own commitments, every recommendation carries a "why this" explanation, notifications can be muted per habit, and prompts fade automatically once a habit is established. The WG challenge shows the shared score of the flat, not who did what, to avoid pressure between flatmates. No dark patterns – artificial scarcity, guilt messages, hidden opt-outs – are used.

G. Testing and Quality Assurance

Testing starts as soon as the first components exist in Sprint 3 and covers three levels: **unit tests** for the R functions (`testthat`), the back-end services (`pytest`) and the iOS view models (XCTest), **integration tests** for the FastAPI–`plumber` bridge and the SMARD ingestion – with recorded API responses as fixtures, so that the tests do not depend on the live service – and **end-to-end tests** of the main user flows from onboarding through recommendation, commitment and check-in, automated with XCUITest on the iOS app. GitHub Actions runs the test suites on every pull request, and Docker keeps the test environment identical for all members. Sprint 5 adds a pilot: the team members use the app for two weeks, which yields the first real check-in data, usability feedback and the before/after measurements needed for the evaluation in Section II-E.

H. Green Software Engineering: the App's Own Footprint

Evaluating and optimising the application's own greenness is a core grading theme of the course, not an extra. The team follows three steps that correspond to the aspects named in the course brief – internal baselines, comparison against common practice, algorithmic efficiency and documented design choices – and draws on the Software Carbon Intensity specification of the Green Software Foundation for the choice of metrics [12].

1) **Baseline (Sprint 4).** The first prototype is measured on the client (Xcode Energy Gauge and Instruments on an iPhone), on the server (CPU time, memory and network bytes per request from Docker statistics and API metrics) and in R (runtime and memory of every analytics job with `bench` and `profvis`). The baseline is documented and targets are set for the following sprints.
2) **Optimise.** Analytics run as nightly batch jobs instead of per request; SMARD data, the daily green-window forecast and the recommendations are computed once and cached; payloads are kept lean and paginated; green-window alerts are scheduled locally on the device and only for windows that match the user's schedule; and algorithmic alternatives for the models in Section II-E are compared by runtime and memory as well as by accuracy.
3) **Prove it.** Every sprint the measurements are repeated, the trend is recorded and shown on the in-app screen "How green is this app?". Each design decision and its measured effect is documented in the final report.

III. SYSTEM ARCHITECTURE

The platform uses a **layered client–server architecture with a separate analytics service** (Fig. 2): a front-end layer (the native iOS app), a back-end layer built with FastAPI, an analytics layer in R and a data layer. All communication between components runs over REST endpoints with JSON payloads; the front-end never talks to R directly. This split is deliberate. The course requires R for all analysis tasks, and isolating R in its own service keeps the general-purpose back-end simple, lets the analytics run as scheduled batch jobs – which is also the most energy-efficient option – and allows the back-end and the analytics to be developed and tested independently by different team members. The style is easy to justify and document in the architecture deliverable of Sprint 2.

A. Frontend Layer

The front-end is a native iOS application written in Swift with SwiftUI [15]. The team compared a web application (React, installable as a progressive web app) with a native iOS app against four criteria – the team's existing skills, notifications, energy measurement tooling and device reach – and chose iOS for three reasons:

- **Prompts are the core mechanism.** The Green Hours planner only works if the alert arrives at the right moment.

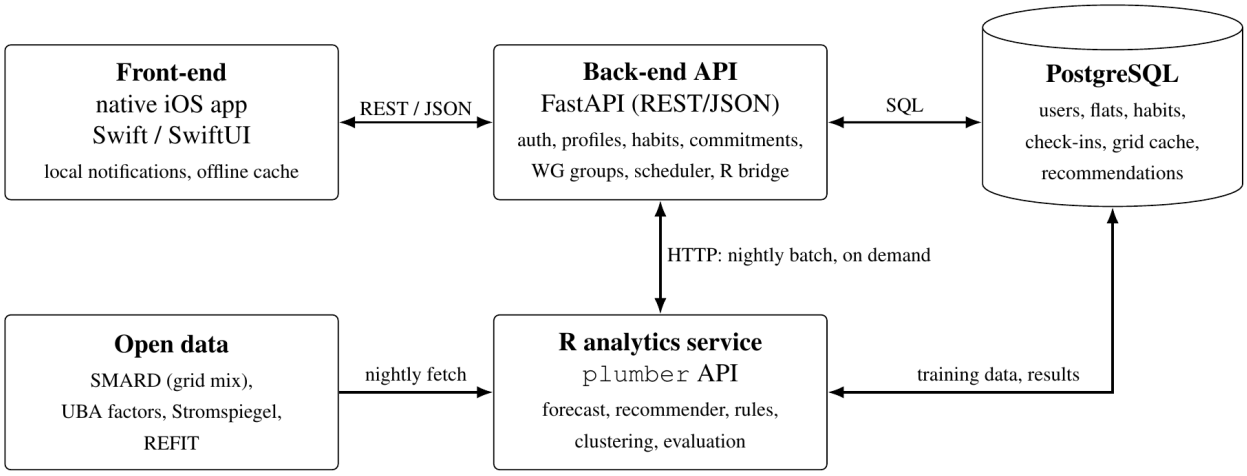


Fig. 2. High-level system architecture: a layered client–server system in which R runs isolated in its own analytics service. All server-side components run as Docker containers.

iOS supports scheduled local notifications natively: after the daily forecast has been fetched, the app schedules the user's green-window alerts on the device, so the server does not have to send a push message for every window.
- **Measurable footprint.** The Xcode Energy Gauge and Instruments break the energy impact of the app down into CPU, network, GPU and other overhead, which feeds directly into the green software evaluation.
- **Existing skills.** Part of the team develops iOS apps professionally, which lowers the risk of the largest user-interface workload in the project.

The main drawback is reach: flatmates with Android phones cannot use the app in this project. Because the app communicates only with the back-end's REST API, whose contract is defined first, a web or Android client can be added later without changes on the server side. The app contains the eleven screens of Section V, caches the current forecast and recommendations so that it also works offline, and is distributed to the pilot users through TestFlight [17].

B. Backend Layer

Implemented with FastAPI [14]. Key components:

- **API gateway** – routing, JWT authentication, request validation.
- **Domain services** – user profiles and flats, the habit catalogue, commitments and check-ins, WG groups, agreements and challenges, notifications.
- **Scheduler** – triggers the nightly SMARD fetch, forecast and recommendation refresh in the R service. Green-window alerts are scheduled on the device itself, so the server sends push notifications only for events such as WG challenge updates.
- **R bridge** – an HTTP client that calls the `plumber` endpoints; results are stored in PostgreSQL so that the API serves cached results instead of recomputing them per request.

C. Analytics Layer

The R analytics service exposes its functions through `plumber` [13]: endpoints for the green-hours forecast, recommendations, personas, association rules and the evaluation reports. It runs in batch mode once per night and on demand only for small requests such as a user's personal green window after a schedule change. The service is stateless, reads from PostgreSQL through a read-only database role (`DBI`/`RPostgres`) and writes its results to dedicated tables. The packages used for the models are chosen and justified in Section II-E.

D. Data Layer

- **PostgreSQL** – users, flats (WGs) and memberships, the habit catalogue, commitments, check-ins, notifications, the hourly grid-intensity cache, recommendations and the green measurements of the app itself.
- **Open data files** – the UBA, Stromspiegel and REFIT extracts are versioned in the repository as CSV files and loaded by the R service.
- **Privacy** – the system stores minimal personal data (e-mail address and display name), uses pseudonymous IDs in the analytics service and offers export and deletion of all personal data in the app.

E. Containerisation and Deployment

The server side is containerised with Docker [16] and orchestrated with Docker Compose. Each major component runs as an isolated service:

- `backend` – Python container running FastAPI with Uvicorn.
- `r_service` – R container exposing the `plumber` endpoints.
- `scheduler` – triggers the nightly jobs and the notification run.
- `postgres` – persistent relational storage.

TABLE VII
TECHNOLOGY STACK

Layer	Technology	Purpose
Front-end	Swift, SwiftUI (native iOS)	User interface, local notifications
Back-end	Python (FastAPI)	REST API, scheduling, R bridge
Analysis	R (`plumber`)	Forecast, recommender, clustering, evaluation
Database	PostgreSQL	Persistent storage, caches
Deployment	Docker Compose, GitHub Actions	Containers, CI
Design	Figma	Wireframes and mockups
Modelling	draw.io / PlantUML	UML diagrams
Management	Jira, GitHub	Backlog, Scrum evidence, code

A single `docker-compose.yml` at the repository root defines all services, networks and volumes, so that every member and the instructor can start the full stack with one command. GitHub Actions builds the images and runs the tests on every pull request. The iOS app is not containerised: it is built and tested with Xcode on a macOS runner and distributed through TestFlight.

IV. TECHNOLOGY STACK

Table VII summarises the technologies per layer. The stack deliberately mirrors the architecture: one general-purpose language for the API, R for everything analytical, and a relational database as the single source of truth. All server-side tools are free, run on macOS, Windows and Linux and are available as Docker images, so that back-end and R work does not depend on a particular machine. iOS development requires a Mac with Xcode, so the iOS area is assigned to members who have one.

V. USER INTERFACES

Eleven interfaces are planned beyond basic CRUD and authentication, three more than the eight required, so that the scope can be reduced without falling below the requirement:

1) **Onboarding energy quiz** – flat size, flatmates, heating type, appliances and weekly schedule, with an immediate "similar flats" comparison.
2) **Energy dashboard** – estimated monthly kWh, CO_2 and euro per end use, and the progress of the current habits.
3) **Recommendation feed with "why this"** – ranked habits, each with the reason it was chosen for this user.
4) **Habit commitment and check-in** – choose a goal, define when and where it happens, log completions, see the streak.
5) **Green Hours planner** – today's and tomorrow's grid CO_2 curve with the user's personal green windows and reminders.
6) **Habit library** – all 22 habits with saving estimates, filterable by type and effort.
7) **One-time checklist** – one-off actions with their lifetime saving.
8) **WG agreement and challenge** – shared goals and house rules of the flat and the ranking between flats.
9) **Trends and analytics** – weekly and monthly development, before/after comparison of adopted habits.
10) **Privacy and data centre** – what is stored, export and deletion, notification settings.
11) **"How green is this app?"** – the measured footprint of the application itself, sprint by sprint.

Wireframes are developed in Figma in Sprint 2 and refined into mockups in Sprint 4.

TABLE VIII
SUCCESS CRITERIA

Area	Criterion and target
Functionality	At least 8 of the 11 planned screens work end-to-end against the live back-end; target: all 11
R analytics	Forecast, recommendation ranking, association rules, clustering and evaluation all run in R in the nightly batch, each covered by `testthat` tests
Green Hours	In a back-test over at least four weeks of SMARD data, the forecast greenest 3-hour window overlaps the actual greenest window on at least 70 % of days
Recommender	In 5-fold cross-validation on synthetic and survey data, the personalised ranking reaches a higher Precision@5 than a popularity baseline
Habit formation	In the pilot, at least two thirds of the participants keep a committed habit with at least 70 % check-in adherence over two weeks; adherence after the prompts fade is reported with a paired test
Usability	Mean System Usability Scale (SUS) score of at least 68, the commonly used average, in the pilot [18]
Green software	Median server CPU time and data transferred per user session at least 30 % lower in the final version than at the Sprint 4 baseline; no model computed per request; "Low" energy impact of the iOS app in a typical session
Process	Every member has been Scrum Master once and Product Owner once; planning, stand-up, review and retrospective evidence in Jira for every sprint; all merged code meets the Definition of Done

VI. SUCCESS CRITERIA

The project is successful if the criteria in Table VIII are met at the final delivery. The quantitative targets are initial values: they are reviewed once the first data and the green baseline are available in Sprint 4, and any change is documented in Jira. Because the pilot involves only the six team members, its results are indicative rather than statistically conclusive.

VII. RISKS AND MITIGATION

Table IX lists the main risks identified during topic selection and how the team intends to handle them. The risk register is kept in Jira and reviewed in every Sprint Retrospective.

VIII. DELIVERABLES

1) Personas, scenarios and user stories
2) Wireframes and mockups (Figma)
3) System architecture document with justification of the architectural style

TABLE IX
RISKS AND MITIGATION

Risk	Mitigation
No real users to train models on	Synthetic users from Stromspiegel, UBA and REFIT; survey in Sprint 2; pilot with the team in Sprint 5
Users cannot measure their own kWh	Estimate from the profile and public statistics; optional meter readings
SMARD API changes or downtime	Nightly caching; fallback to the stored dataset; ingestion isolated in one R module
R skills uneven across the team	Two R owners; shared R scripts; pair-programming sessions
Nudges feel manipulative or annoying	User-chosen commitments, mute option, "why this" on every tip, prompts fade once a habit is formed
iOS-only app excludes flatmates with Android phones	Client-independent REST API defined first, so a web or Android client can be added later; Android support listed as future work
Scope too large for one semester	Eight screens required, eleven planned; the additional screens are the first candidates to be descoped

4) UML models: use case, class, activity and sequence diagrams
5) Working native iOS app with back-end and R analytics service; the server side containerised with Docker
6) Green software evaluation: baseline, optimisations and measured trend of the app's own footprint
7) Jira project with Scrum evidence, artifacts and team contract, accessible to the instructor
8) User documentation and final project report

IX. CONCLUSION

Home Energy Coach addresses a group that most energy advice ignores: students and renters in shared flats, who cannot renovate their building but can change when and how they use energy. The proposal combines public data on the German electricity grid with a short user profile to recommend habits that fit each flat and each schedule, and uses established behaviour-change principles – small commitments, prompts at the right moment and immediate feedback – to turn them into routines. The architecture keeps all analysis in a separate R service, as the course requires, and runs it as nightly batch jobs, which also keeps the app's own footprint low. With a defined scope, measurable success criteria, a role rotation for all six members and a pilot on the team itself, the plan fits the five sprints and will give the team concrete evidence at the final delivery on both the effect of the recommender and the greenness of the software.

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